



An improved CNN-LSTM model-based state-of-health estimation approach for lithium-ion batteries

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ABSTRACT

Accurate SOH (State of Health) estimation is one of the key technologies to ensure the safe operation of lithium-ion batteries. When predicting SOH, efficient data feature extraction is the premise to ensure accurate prediction. In this work, a feature selection method is proposed to help neural networks train more effectively by removing useless features from the input data during the data preparation step. In addition, the skip connection is added to the convolutional neural network-long short-term memory (CNN-LSTM) model in this work to address the problem of neural network degradation caused by multi-layer LSTM. The presented approach is validated on the NASA and Oxford battery dataset. The results demonstrate that after using the feature selection approach to remove the less significant features, the SOH prediction accuracy is enhanced and the computational load on the neural network is decreased. Compared with other neural network models, the CNN-LSTM-Skip model has better robustness and higher accuracy under different conditions, and the RMSE is below 0.004 on the NASA dataset and the Oxford dataset.

1. Introduction

Lithium-ion batteries have excellent characteristics such as low self-discharge rate, high power density, no memory effect, and long cycle life. Therefore, in the fields of consumer electronics and new energy vehicles, lithium-ion batteries are widely used as energy storage components [1]. Lithium-ion batteries must be equipped with a battery management system (BMS) to increase their lifespan and make sure they are operating in a secure environment [2]. The main goal of the BMS is to operate the battery within a safe range by monitoring parameters such as state of health (SOH) and state of charge (SOC). As one of the key functions of BMS, accurate SOH estimation is crucial to ensure the driving experience and safe driving of electric vehicles [3]. SOH is an indicator of the battery life, which reflects the aging degree and reliability of the battery. It is usually defined as the ratio of the current capacity to the nominal capacity in the literature.

Different methods are deployed to reach an accurate battery SOH estimation. These methods can be listed in three categories: direct measurement methods, model-based methods, and data-driven methods [4]. Direct measurements estimate the value of SOH by measuring parameters directly related to battery degradation, such as open circuit

voltage (OCV), internal resistance, and impedance [5]. Although the computational complexity of the method is not high and the method has strong adaptability to different batteries, it is difficult to apply under online operating conditions because of the strict hardware requirements [6]. Among the model-based prediction methods, electrochemical models and equivalent circuit models are commonly used. Electrochemical models use complex partial differential equations to describe the internal physical and chemical mechanisms of battery degradation, such as pseudo-two-dimensional models (P2D models), which require a large amount of calculation [7]. In practical applications, some parameters of the partial differential equation cannot be measured, and the electrochemical model is difficult to be applied [8]. The equivalent circuit model (ECM) builds a circuit model composed of electrical components, which is equivalent or approximate to the dynamic characteristics of the battery system [9]. Compared to other methods, ECMs are much easier to approximate the characteristic of the battery, while there are difficulties in the identification of parameters under different conditions [10].

The battery degradation process involves complex chemical mechanisms, showing a high degree of nonlinearity [11]. Additionally, due to the complexity of the electrochemical mechanism in the battery, the SOH estimation of different batteries needs to be remodeled [12].

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Acronyms			
BMS	battery management system	$\mathbf{x}$	input feature vector
SOH	state of health	$\mathbf{X}$	input matrix
SOC	state of charge	$\mathbf{E}$	the error matrix
OCV	open circuit voltage	$\mathbf{A}$	the undetermined coefficient matrix of SG filter
P2D model	pseudo-two-dimensional models	$\hat{\mathbf{Y}}$	the result of SG filter
ECM	equivalent circuit model	L	loss function
PCC	Pearson correlation coefficient	$\tanh$	$\tanh$ activation function
NCA	neighborhood component analysis	σ	sigmoid activation function
CNN	convolutional neural network	$\mathbf{x}^t$	feature contains information
LSTM	long short-term memory	$\mathbf{x}^r$	feature without information
SVR	support vector regression	f	Trained model
IC	incremental capacity	$e(f)$	the loss function value trained in model f
PCA	principal component analysis	$\mathbf{K}_i$	the i -th convolution kernel
CNN-LSTM-Skip	CNN-LSTM model with Skip connection	$\mathbf{S}^l$	the feature map of the l -th layer
MAE	mean absolute error	w	the width of the pooling area
RMSE	root mean square error	$\mathbf{W}$	the weight matrix
R^2	coefficient of determination	$\mathbf{b}$	the bias vector
ARMSE	average root mean square error	$\mathbf{f}_t$	the result of the forget gate
EIS	electrochemical impedance spectroscopy	$\mathbf{h}_t$	the output of a LSTM cell
Nomenclatures		$\mathbf{i}_t$	the output of the input gate
Q	capacity(Ah)	$\mathbf{c}_t$	the current cell state
t	time(s)	$\mathbf{o}_t$	the result of the output gate
V	voltage(V)	C_0	the rated capacity of the battery
I	current(A)	Q_{max}	the current maximum available capacity of the battery
T	temperature(CB)	e_{origin}	the original error of the model
N	length of the SG filter window	$\mathbf{X}_j$	input matrix with no information of feature j
		e_j	the model trained with $\mathbf{X}_j$ error
		I_{pj}	the importance of feature j

Recently, studies based on data-driven methods have attracted a lot of interest due to their flexibility and universality. The data-driven model relies simply on battery operating data, such as current, voltage, and temperature. Typical data-driven models include Gaussian process regression [13], support vector machine [14], extreme learning machine [15], grey correlation analysis [16], and neural network [17], which directly analyze the relationship between input data and SOH without analyzing the battery aging mechanism. Data-driven methods are mainly divided into the following steps: data collection and pre-processing, feature extraction and selection, data-driven modeling, validation, and testing [18,19]. Neural networks are able to learn and adapt to complex patterns in data, making them well-suited for tasks such as predicting the SOH of a system based on a variety of input data [17,20].

In the data-driven model, it is critical to select the appropriate health indicator (HI) as input. In Ref. [21], fifteen health indicators were extracted from the battery charging and discharging process, and the health indicators were selected by Pearson correlation coefficient (PCC) and neighborhood component analysis (NCA) methods. Based on long short-term memory (LSTM) combined with differential evolution grey wolf optimizer (DEGWO), the global search ability of the model was improved, and the estimation performance was improved. Liu et al. extracted 9 features from the current curve in the constant voltage charging process and used the Pearson correlation coefficient to select the features [22]. They proposed a multi-feature fusion model to estimate battery SOH by fusing different feature parameters and combining support vector regression (SVR) and long short-term memory network (LSTM). Li et al. manually extracted the health features from the incremental capacity (IC) curves of the partial incremental analysis and used the SVR-based battery degradation model for SOH prediction and these features showed excellent predictive ability [23]. Che et al. combined the health indicators with sequential information highly correlated with battery capacity and life and used the Gaussian mixture

model for life cluster analysis to verify the validity of the health indicators [19]. Li et al. established an end-to-end framework based on an improved Bayesian hyperparameter optimization algorithm to reduce the workload of hyperparameter tuning. The framework was based on the CNN-ASTLSTM model, using the original data of long sequences such as current and voltage, and achieved a lower root mean square error while reducing the amount of calculation [24]. Gu et al. used PCC to analyze feature correlation and select features and used the principal component analysis (PCA) method for feature dimensionality reduction [25]. They proposed a CNN-Transformer model which used CNN local features coupled with Transformer-based global properties to improve the accuracy of SOH prediction.

It takes a long time to train the neural network model to get the expected results, so it is critical to limit the number of input features of the model and the dimension of the dataset. In the previous literature, the research on the importance of features is still limited. PCC is mainly used for feature selection, which only judges the linear correlation of features, while the neural network model is highly nonlinear. Using the PCC method may ignore some nonlinear features. This work aims to improve the accuracy of the prediction result of the neural network model and reduce the computational burden of the neural network model. The main contributions are summarized as follows: (1) The incremental capacity curve is used as the input of the neural network model, and the CNN network is used to extract features. Adding an input IC curve to the neural network can capture more features and improve prediction accuracy. (2) Based on the variable importance algorithm, the validity of each input feature is calculated, and the invalid features are deleted. In the process of calculating the validity of the indicators, the importance of the IC curve is verified. (3) This work presents a CNN-LSTM model with Skip connection (CNN-LSTM-Skip) to overcome the neural network model degradation problem caused by multi-layer stacked LSTM [26].

The work is organized as follows: Section 2 describes the structure of

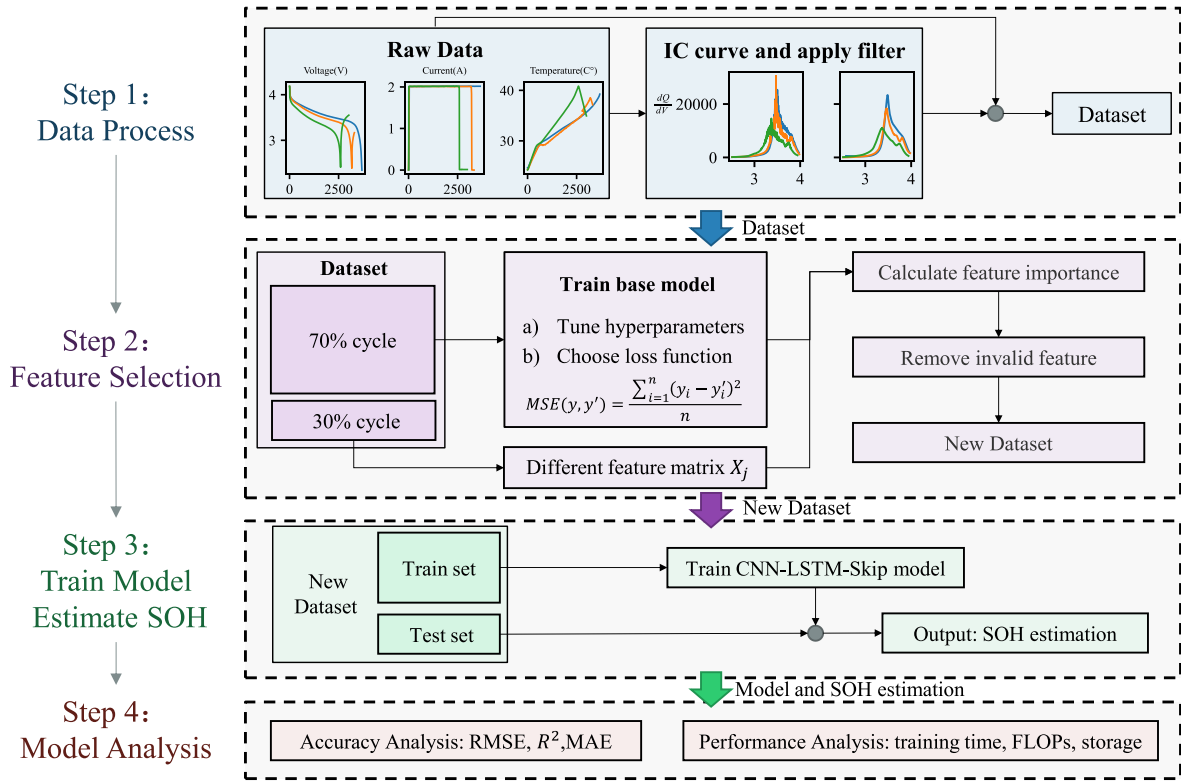


Fig. 1. Battery prediction framework based on CNN-LSTM-Skip model.

the variable importance algorithm and the CNN-LSTM-Skip approach. Section 3 shows the specific process of data preprocessing and uses the variable importance algorithm to filter the features. Section 4 compares the accuracy and performance of CNN-LSTM, CNN-ASTLSTM, and CNN-LSTM-Skip. Section 5 concludes this study.

2. Methodology

The model framework is presented in Fig. 1 and is comprised of four steps. In the data processing stage, to make full use of the information of the original battery data, the IC curve is added to the model input. In the feature selection stage, to reduce the computational burden of the neural network model and judge the importance of features to the output of the neural network more effectively, a variable importance algorithm is used to remove invalid features in the input data, and the importance of the IC curve is verified. Finally, the CNN-LSTM-Skip model is proposed to tackle the neural network degradation problem.

2.1. Incremental capacity analysis

In recent years, the Incremental Capacity (IC) analysis method has been regarded as an important method for estimating SOH. The key to this method is applying differential equations to derive the battery's IC curve, which is determined from the charging or discharging process under a constant current condition [27].

The relationship between battery capacity Q and battery voltage V under constant current discharge conditions is as follows:

$$Q = ItV \quad (1)$$

where I is constant current and t is time. There is a functional relationship between Q and V , and the formula for the IC curve can be deduced as:

$$f(V) = \frac{dQ}{dV} = \frac{I \cdot dt}{dV} = I \cdot \frac{dt}{dV} \quad (2)$$

The formula used in discrete sensor data using is:

$$\frac{\Delta Q}{\Delta V} = \frac{Q_{V_k|V_{k+1}} - Q_{V_{k+1}|V_k}}{V_{k+1} - V_k} = \frac{I(t_{k+1} - t_k)}{V_{k+1} - V_k} \quad (3)$$

where $Q_{V_k|V_{k+1}}$ represents the capacity change when the voltage changes from V_k to V_{k+1} , k represents the number of sampling times, and t represents the sampling time.

According to Equation (3), the IC curve can be described as the differential of capacity to voltage during constant current charging or constant current discharging. The value of peaks and peak positions in IC curves are key features that many battery SOH estimation models often use and achieve good results [28].

IC curves are sensitive to noise, so the original curve usually contains many signal interferences that effective features cannot be extracted directly. Therefore, it is also necessary to smooth the IC curve with a filter. The Savitzky-Golay (SG) filter is used in this study, which is widely used in data smoothing and denoising [29]. Its core idea is to perform weighted filtering on the data in the window, and its weight is obtained by least squares fitting through a given high-order polynomial [30]. The basic principle of filter is as follows:

$$N = 2m + 1 \quad (4)$$

$$\mathbf{x} = [x_{-m}, x_{-m+1}, \dots, x_{m-1}, x_m] \quad (5)$$

$$y_i = a_0 + a_1x_i + a_2x_i^2 + \dots + a_{k-1}x_i^{k-1} \quad (6)$$

where N is the length of the window, m is the number of data points on the left and right sides of the fitting point y_i in a window, and x is the data points in the window.

Polynomials in multiple windows can be written in matrix form as follows:

$$\mathbf{Y}_{(2m+1) \times 1} = \mathbf{X}_{(2m+1) \times k} \mathbf{A}_{k \times 1} + \mathbf{E}_{(2m+1) \times 1} \quad (7)$$

where $\mathbf{A}$ represents the undetermined coefficient matrix, $\mathbf{Y}$ represents

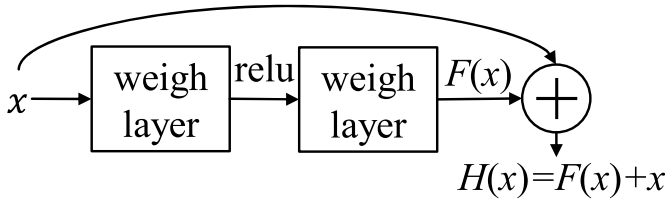


Fig. 2. Skip connection.

the fitting point matrix, E represents the error between the fitting point and the data point, X represents the data point matrix, and the subscript is the shape of the matrix.

The undetermined coefficient matrix A can be solved by the least square method:

$$\hat{A} = (X^T \cdot X)^{-1} \cdot X^T \cdot Y \quad (8)$$

where $\hat{A}$ is the least square solution.

Thus, the smoothed result of $\hat{Y}$ can be given by

$$\hat{Y} = X \cdot \hat{A} = X \cdot (X^T \cdot X)^{-1} \cdot X^T \cdot Y \quad (9)$$

2.2. Feature selection algorithm

The goal of the feature importance algorithm is to measure the dependence of the model on the feature. To illustrate the algorithm in detail, the following variables are now introduced. Suppose a neural network model that has been trained is f , and the model f is trained based on two features x_1 and x_2 , and the feature data are all normalized. The evaluation result of the model on the original dataset is $e_t(f) = L(f, (Y, x_1^t, x_2^t))$, L is a loss function. RMSE is adopted here. Then the x_1^t features in the training set are replaced with the random series x_1^r without information. Calculated from the model here:

$$e_1(f) = L(f, (Y, x_1^r, x_2^t)) \quad (10)$$

where $e_1(f)$ represents the loss function value trained in model f using x_1^r , x_2^t . The importance (I_p) of feature x_1 in model f can be defined as follows:

$$I_{p1}(f) = e_1(f) - e_t(f) \quad (11)$$

Equation (11) is based on the differential definition [31], which lacks interpretation. But it has several advantages in computation. When X_1 is a random variable, the mean, variance, and asymptotic distribution of the estimator $\widehat{I_{p1}(f)}$ can be determined by the results of the U statistic without using the Delta method [32].

$$I_{p1}(f) = e_1(f) / e_t(f) \quad (12)$$

Equation (12) is defined in terms of ratios [33], which allow comparisons in many different problems regardless of the dimension. The larger the value of $I_{p1}(f)$, the higher the importance of x_1 in the model f . When the value of $I_{p1}(f)$ is greater than 10, it indicates that the model f is heavily dependent on x_1 . When the value of $I_{p1}(f)$ is 1, it means that the feature has no contribution to the output of the neural network. There is also the possibility that $I_{p1}(f)$ is less than 1, which indicates that model f incorrectly uses the information in x_1 to output a biased result or misclassifies. In this case, for model f , there is a better model f' that satisfies $I_{p1}(f') \geq 1$ and $e_t(f') < e_t(f)$, which means that the model f is a suboptimal model. In this work, the neural network model is used. For the convenience of comparison, the importance I adopts the definition method of formula Equation (12).

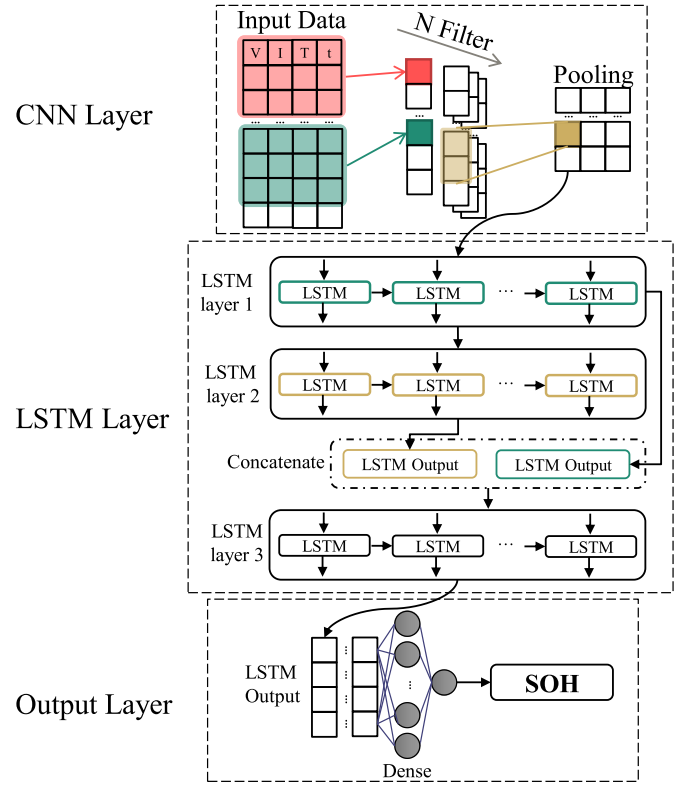


Fig. 3. CNN-LSTM with skip connection model.

2.3. CNN-LSTM with skip connection model

When multiple layers in the neural network are stacked, there will be a degradation problem, that is, as the depth of the neural network increases, the accuracy rate tends to saturate, and then declines, bringing more errors to the prediction results. He et al. added a residual link mechanism to Resnet [34], as shown in Fig. 2. $F(x)$ is a nonlinear change function, x represents the input of the network, and $H(x)$ represents the output of the residual connection part. The input x makes the output optimal, and then changing the feature x through any transformation will make the loss larger, $F(x)$ will automatically tend to 0, and x will not change and continue to pass along the original path. This achieves the purpose without increasing the computational cost: in the forward process, when the output of the layer has been optimized, the network layer behind the deep network achieves the role of constant mapping. The skip-connection is implemented in another way in DenseNet [35], concatenating the outputs of different layers instead of using addition. This work uses the method in DenseNet.

Fig. 3 shows the structure of the CNN-LSTM-Skip. The preprocessed data first enters a one-dimensional convolutional layer, which extracts features by moving the filter along the time axis. The process of feature extraction is as follows:

$$S^l = f \left(\sum_{i=1}^{N_{CK}} X_0 * K_i + b_i^l \right) \quad (13)$$

where X_0 is the input data, K_i is the i -th convolution kernel, S^l is the feature map of the l -th layer, $f(\cdot)$ is the activation function, N_{CK} is the number of convolution kernels used in the convolution layer, b_i^l is the i -th bias of the l -th feature map, $*$ is the convolution operation. The activation function is the Relu function.

By the pooling operation, the length of input data for the next layer is reduced obviously, which is helpful to alleviate the over-fitting phenomenon in the training process. In practice, the max-pooling is

commonly used, and the maximum value in the pooling area is used as the output. The max-pooling formula is as follows:

$$S^l(j) = \max_{(j-1)w+1 \leq t \leq jw} \{X^{l-1}(t)\} \quad (14)$$

where $S^l(j)$ is the output of the j -th pooling area of the l -th layer, w indicates the width of the pooling area, $X^{l-1}(t)$ is the pooling area.

After using CNN to extract features from the original data, LSTM is used to predict SOH. Recurrent Neural Networks (RNNs) suffer from the vanishing gradient problem, and LSTM is designed to overcome this problem by being able to learn both short-term and long-term dependency. The information is transmitted through the forget gate, the input gate, and the output gate, thus controlling the degree of memory and forget of previous and current information.

The forget gate determines the invalid information forgotten by the forget unit. Input the output value of the previous moment and the input value of the current moment into the forget gate, and calculate the output value of the forget gate, as shown in the following formula:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (15)$$

where f_t is the result of the forget gate, W_f is the weight of the forget gate, b_f is the bias of the forget gate, x_t is the input value at the current moment, h_{t-1} is the output of the previous LSTM cell.

The output value of the last time step and the input value of the current time step are input into the input gate, and the output value and candidate cell state of the input gate are obtained after calculation, as shown in the following formulas:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (16)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (17)$$

where i_t is the output of the input gate, W_i is the weight of the input gate, b_i is the bias of the input gate, W_c is the weight of the candidate input gate, and b_c is the bias of the candidate input gate.

Update the current cell state as follows:

$$c_t = f_t \times c_{t-1} + i_t \times \tilde{c}_t \quad (18)$$

The output h_{t-1} and input x_t are received as input values of the output gate at time t , and the output o_t of the output gate is obtained as follows:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (19)$$

where W_o is the weight of the output gate, and b_o is the bias of the output gate.

The output value of LSTM is obtained by calculating the output of the output gate and the state of the cell, as shown in the following formula:

$$h_t = o_t \times \tanh(c_t) \quad (20)$$

Finally, the fully connected layer is used to output the optimal result.

3. Numerical experiments

3.1. Dataset description

3.1.1. NASA dataset

We employ the dataset provided by NASA PCoE, which uses 18650 lithium-ion batteries with rated capacity of 2Ah [36]. The battery underwent three different operations during each cycle of the experiment, including charging, discharging, and electrochemical impedance spectroscopy (EIS). In the charging stage, the battery is charged with a constant current of 1.5 A first, charged with a constant voltage of 4.2 V when the battery voltage is 4.2 V, and stops when the charging current is reduced to 20 mA. In the discharge stage, the battery stops when it discharges to the set cut-off voltage. In the impedance spectroscopy measurement stage, the electrochemical impedance measurement is

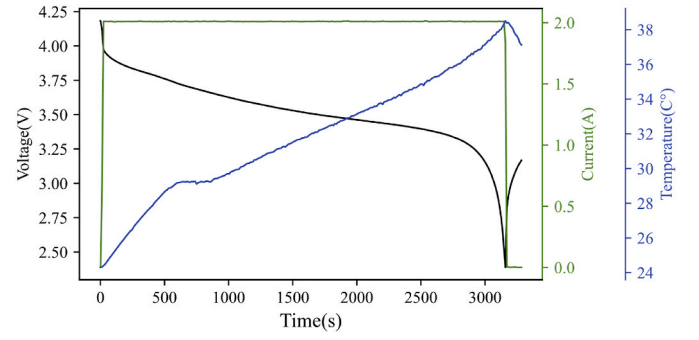


Fig. 4. The discharge data of the NASA battery.

used to measure the impedance of the lithium-ion battery at a frequency from 0.1 Hz to 5.0 kHz. The aging test ends when the battery capacity drops to 70% of the rated capacity.

Voltage, current, and temperature data were collected during the cycle, as shown in Fig. 4. Four batteries cycled under conventional conditions in the dataset were selected, and the numbers are B0005, B0006, B0007, B0018. The discharge cut-off voltage of these 4 batteries is 2.5 V, 2.7 V, and 2.2 V during the discharge test. The constant current discharge stage of the battery data is selected.

3.1.2. Oxford dataset

The Oxford dataset contains aging data for 8 Kokam lithium-ion batteries with a rated capacity of 740mAh [37]. At an ambient temperature of 40 °C, use the CC-CV charging mode and the ARTEMIS urban driving cycle discharge mode to repeatedly discharge and recharge 8 test lithium-ion batteries at a constant current of 2C (1.48 A). After every 100 cycles, a characterization test is performed. The characterization measurement is 1C constant current charge and discharge. The data sampling frequency is 1 Hz. The data is shown in Fig. 5. The data of 8 batteries discharged at constant currents are selected, including voltage, current, sampling time, temperature, and accumulated capacity.

3.2. Data preprocessing

3.2.1. Extracting and smoothing IC curves

Incremental capacity analysis is a common method for battery state-of-health prediction. The IC curve contains a lot of features, such as peak height, peak position, peak area, peak width, and left and right slopes of the peak, etc., which can provide key information about the aging mechanism and SOH of the battery. The original IC curve is shown in Fig. 6(a). Since the curve is sensitive to noise, the noise contained in the original curve may cover up the effective features, and it is difficult to be directly used for health state estimation.

IC curve needs to use a filtering algorithm to remove noise, commonly used algorithms include wavelet transform filtering, Gaussian filtering, and Savitzky-Golay filter. Different filter algorithms have different smoothing effects, and the filtering algorithm may destroy the original information of the data. The most important feature of Savitzky-Golay filter is that it can ensure the shape and width of the signal remains unchanged while filtering out the noise, and this algorithm can preserve the low frequency signal data while removing the high frequency part [38]. Therefore, the SG filter is used in this work, and reasonable parameter settings are used to ensure that the characteristics of the IC curve are not destroyed. The SG filter window length is 41 and the polynomial fitting order is 4. After filtering, as shown in Fig. 6 (b), it can be observed that the IC curve changes with the number of cycles, and as the number of battery cycles increases, the peak of the IC curve decreases, and the peak position appears earlier.

3.2.2. SOH definition

SOH is an indicator of battery performance degradation, and the

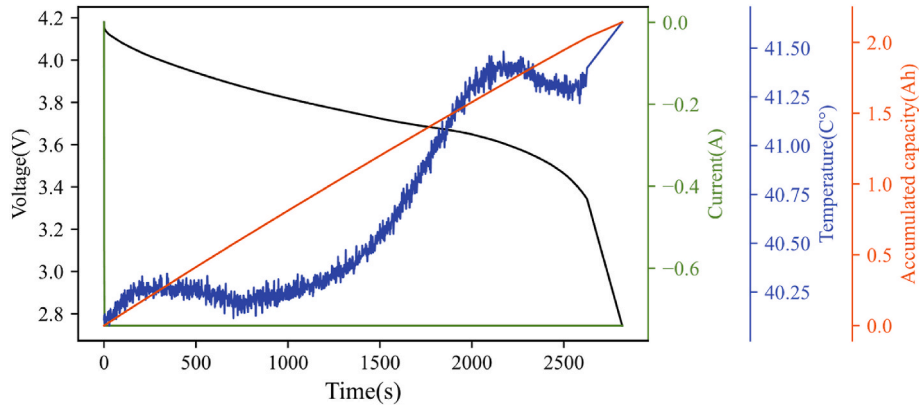


Fig. 5. The discharge data of the Oxford battery.

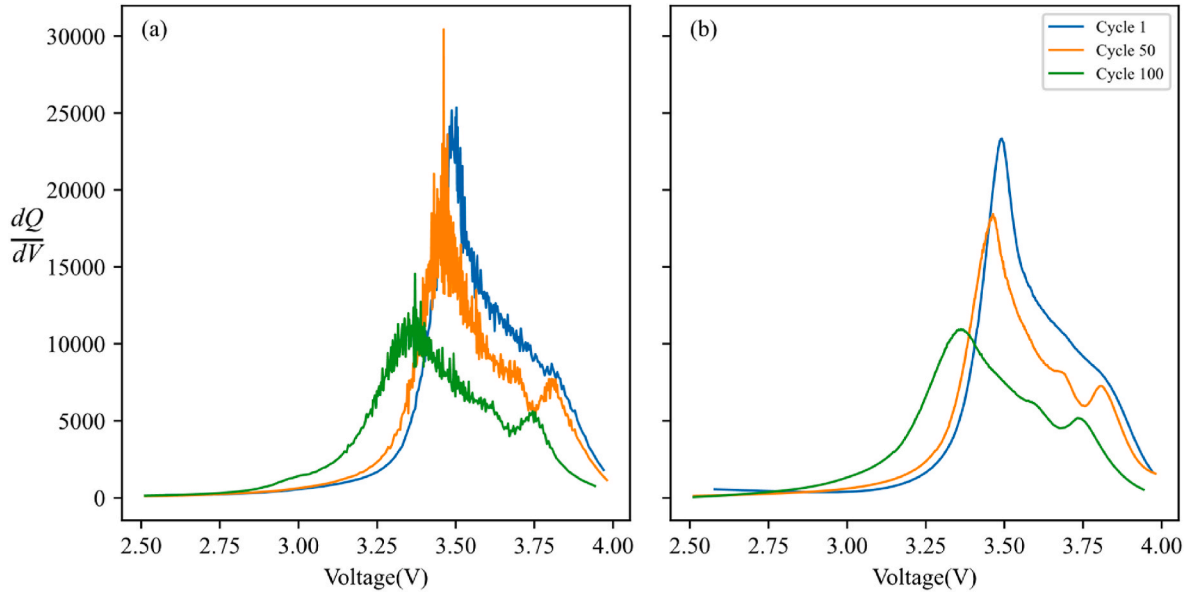


Fig. 6. NASA B0006 IC curve.

aging degree of the battery can be measured according to SOH. SOH refers to the percentage of the current battery capacity to the initial capacity, which is defined as follows [39]:

$$\text{SOH} = \frac{Q_{\max}}{C_0} \times 100\% \quad (21)$$

where C_0 is the rated capacity of the battery and $Q_{\max}$ is the current maximum available capacity of the battery. If the SOH of the battery is below 80%, it indicates that the battery needs to be replaced.

3.2.3. Data normalization

Data normalization is a conventional way of data processing in the field of deep learning. Normalization can help to stabilize the gradient descent step, allowing us to use larger learning rates or help models converge faster for a given learning rate. The formula for data normalization is as follows:

$$x^* = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (22)$$

where x^* is the normalized data, x_i is the original data, $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the original data respectively.

3.3. Feature importance calculation and feature selection

To verify that the input of the IC curve is helpful for the prediction of the neural network, and to delete invalid input data, the second step of the experiment is to determine the validity of each input feature. Here, the method of Fisher [33] was used. The flow of the algorithm is described below:

Step 1. Train the base model f , determine the input features $\mathbf{X}$, output target $\mathbf{y}$, and the error calculation method L is RMSE.

Step 2. The original error of the model e_{origin} is calculated as follows:

$$e_{\text{origin}} = L(\mathbf{y}, f(\mathbf{X})) \quad (23)$$

Step 3. For each feature $j \in \{1, \dots, p\}$ in the original input matrix $\mathbf{X}$, select feature j , disrupt the order of feature j , so that feature j does not contain valid information, and generate a new feature matrix $\mathbf{X}_j$. After this processing, the result $\mathbf{y}_j$ generated with matrix $\mathbf{X}_j$ is uncorrelated with feature j . Next, the error e_j can be calculated as:

$$e_j = L(\mathbf{y}, f(\mathbf{X}_j)) \quad (24)$$

Then calculate the importance of feature j :

Table 1
RMSE results for the Nasa dataset.

Feature	Battery ID				Average RMSE
	B0005	B0006	B0007	B0018	
Voltage	0.1318	0.1099	0.1085	0.1503	0.1251
Current	0.0471	0.0467	0.0385	0.0488	0.0453
Sampling time	0.0889	0.0703	0.0725	0.0926	0.0811
Temperature	0.0394	0.0381	0.0462	0.0307	0.0386
IC curve	0.1085	0.1015	0.1444	0.1481	0.1256
Base model	0.0133	0.0195	0.0102	0.0098	0.0132

$$I_{pj} = \frac{e_j}{e_{\text{origin}}} \quad (25)$$

Finally, sort the calculation results. The greater the importance of feature j to the output of the neural network, the greater the difference between the prediction result and the original error using the new feature matrix $\mathbf{X}_j$, and the greater the importance of feature j .

The NASA dataset contains current, voltage, time, and temperature data, while the Oxford dataset contains data on current, voltage, time, temperature, and accumulated charge/discharge capacity data. Thus, we choose all the data as the input features. During this experiment, the RMSE of number 4 batteries was collected for each feature. The RMSE in Table 1 and Table 2 is the average result after using 5 fixed random number seeds. The ARMSE is the average value of the RMSE of each battery, and ARMSE is used in the final calculation of the importance.

Table 1 shows the experimental data of the feature ranking algorithm on the NASA dataset, and the calculation results of the importance index I_p are shown in Fig. 7. The base model does not perform feature replacement. First, comparing the ARMSE of the base model and the model with the IC curve feature replaced, the IC curve contributes significantly to the output of the neural network. Secondly, according to the indicator I_p , the voltage, sampling time, and IC curve contribute more to the results of the neural network, while the discharge current and temperature contribute less. Based on the above results, two redundant features of current and temperature were removed.

Table 2 shows the ARMSE calculation results of the Oxford dataset. From Fig. 7, the four features, time, accumulated capacity, voltage, and IC curve, contribute significantly to the neural network output results, thus discarding the current and the temperature features.

In this section, to remove redundant features, a feature importance calculating algorithm was used to calculate the contribution of each feature to the output of the neural network based on the NASA dataset and the Oxford dataset. After feature selection, the amount of data input by the neural network is 60% and 80% of the original, and the

calculation speed is accelerated.

4. Model comparison

To verify the proposed neural network model, the CNN-LSTM model [40] and CNN-ASTLSTM model [24] are selected to compare with CNN-LSTM-Skip. Both CNN-LSTM and CNN-ASTLSTM have two layers of LSTM. During the experiment, 60%, 70%, and 80% of the cycle data were used for model training, while the remaining 40%, 30%, and 20% of the cycle data were used for testing. We conduct an experiment on the NASA dataset and the Oxford dataset. The hardware and software configurations used are: Intel(R) Core (TM) i5-6500, 16G Memory, NVIDIA GeForce GTX 750Ti, Python 3.9, Tensorflow-GPU 2.6.0, Keras 2.6.0.

The detailed structure of the CNN-LSTM-Skip neural network is shown in Table 3. The hyperparameters of the model trained on the NASA dataset are shown in Table 4.

Table 3
Detailed structure of CNN-LSTM-Skip model.

Layer	Activation function	Output data shape
Input	N/A	None, 300, 3
CNN 1D	relu	None, 38, 50
MaxPooling1D	N/A	None, 19, 50
LSTM1	tanh	None, 19, 50
Dropout	N/A	None, 19, 50
LSTM2	tanh	None, 19, 50
Concatenate (LSTM1,LSTM2)	N/A	None, 38, 50
Batch Normalization	N/A	None, 38, 50
LSTM3	tanh	None, 38, 50
Dense	N/A	None, 38, 1

Table 4
Model hyperparameters.

	CNN- LSTM	CNN-ASTLSTM	CNN-LSTM-Skip
Batch size	20	20	20
Dropout	0.0214	0.0609	0.0214
filters	50	50	50
strides	8	4	8
Pool size	2	2	2
Kernel size	8	8	8
learning rate	0.0003	0.0004	0.0003
epoch	200	200	200
LSTM1	50	50	50
LSTM2	50	50	50
LSTM3	N/A	N/A	50

Table 2
ARMSE calculation results for the Oxford dataset.

Feature	Voltage	Current	Time	Temperature	IC curve	Accumulated capacity	Base model
ARMSE	0.0517	0.0114	0.1558	0.0113	0.0577	0.0946	0.0079

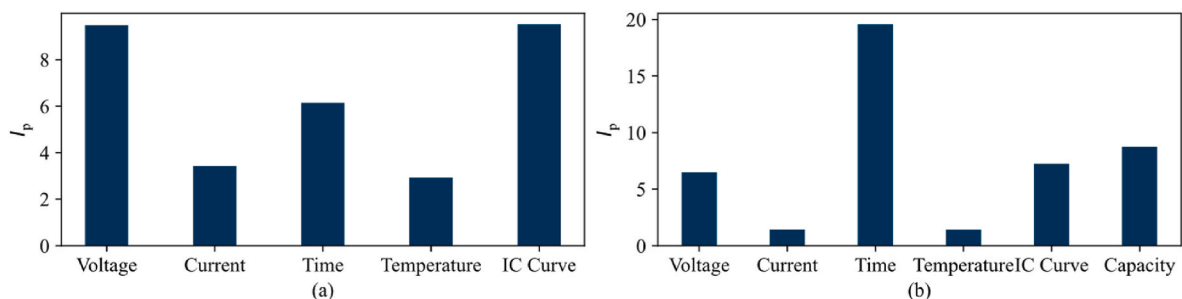


Fig. 7. Importance I_p calculation results.

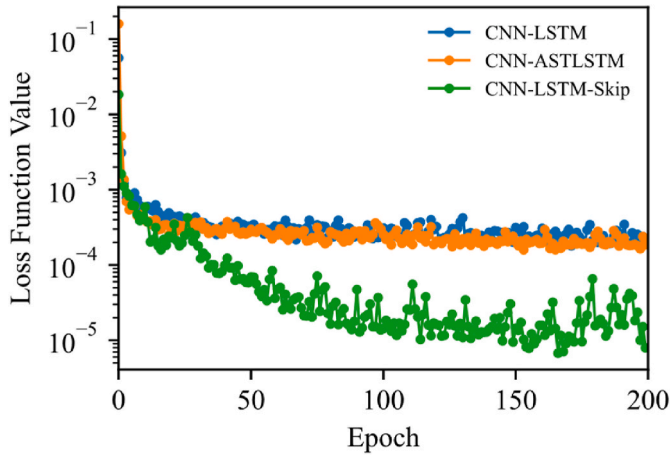


Fig. 8. Loss function value in the model training process.

4.1. Evaluation indicators

To evaluate the performance of the proposed model, we evaluate the proposed model from two perspectives. From the perspective of accuracy, the mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination R^2 are used. The mean absolute error is used to measure the error between the predicted value and the true value. The smaller the MAE, the better the model and the weight of all errors is the same. RMSE is a typical indicator of a regression model, which is used to indicate how much error the model will make in the prediction, and for larger errors, the weight is higher. The coefficient of determination is used to evaluate the goodness of fit and regression. The closer the value of R^2 is to 1, the better the fitting effect is, and the smaller the value is, the worse the fitting effect is.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (26)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (27)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (28)$$

$$ARMSE = \frac{1}{n} \sum_{i=1}^n RMSE_i \quad (29)$$

where y_i , $\hat{y}_i$, $\bar{y}$, $RMSE_i$ represent the actual value, the predicted value of the neural network, the average of the actual value, and the RMSE of the i -th cell, respectively.

To measure the computer resource occupancy of the model, the training time, FLOPs (floating point operations), number of training parameters, and the model storage space were recorded during the training of the neural network model. Training time refers to the time to train the neural network using the training set. FLOPs refer to the number of floating-point operations, understood as the amount of calculation, which can be used to measure the complexity of the algorithm/model and is exported by the model analysis module of Keras. The number of parameters indicates the size of the neural network, which is derived using “model.summary” method. The size of the model storage space refers to the size of the H5DF file generated using the Keras built-in save module, which contains the structure of the neural network and the parameters after training.

Table 5
Statistics for the loss function.

	CNN-LSTM	CNN-ASTLSTM	CNN-LSTM-Skip
Variance	0.000016	0.000042	0.000002
Mean	0.000607	0.000435	0.000180

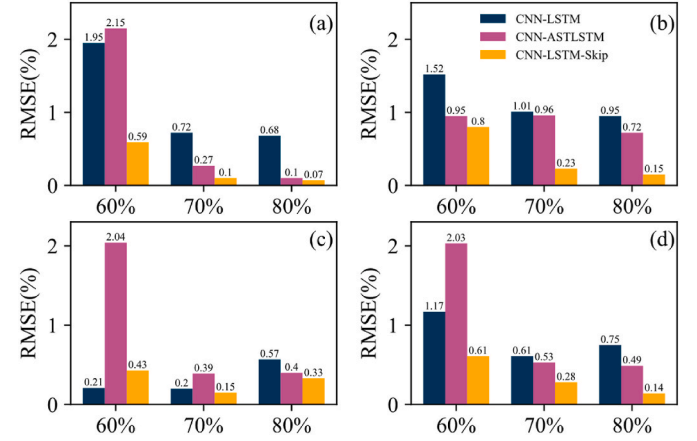


Fig. 9. The SOH estimation result of the model trained with different data volumes, (a) B0005, (b) B0006, (c) B0007, (d) B0018.

4.2. Accuracy comparison

4.2.1. The experience with data from NASA

The batteries numbered B0005, B0006, B0007, and B0018 in the NASA dataset were used for experiments.

Fig. 8 shows how the loss function changes as the neural network are trained using 60% of the cycle data. To clearly show the change in the final loss function value, the y-axis adopts a logarithmic coordinate, which is non-linear, and fluctuations in small values are amplified. Table 5 shows the mean and variance of the three models. The variance data shows that the proposed model has the smallest variance and the smallest fluctuation. The loss function of the training neural network adopts mean square error. The smaller the loss function, the higher the prediction accuracy of the model on the training set. From the final result of the loss function, it can be seen that the CNN-LSTM-Skip model has the highest prediction accuracy. CNN-LSTM and CNN-ASTLSTM are close, but CNN-ASTLSTM performs better than CNN-LSTM in the comparison of subsequent results.

Fig. 9 compares the RMSE of the three models when using different data volumes for training, and the x-coordinate is the percentage of the training data volume in the total cycle data. It can be seen from Fig. 9 that as the training data volume increases, the accuracy of the three models will increase, and the CNN-LSTM-Skip model has the highest accuracy when trained with different amounts of data. As shown in Fig. 10(a)~(c), the model proposed in this work has the best accuracy when using 60% of the data training. The accuracy of the three models is relatively similar when trained with 70% and 80% of the cycle data. To clearly show the accuracy of the model estimation results and the error distribution, the model error box diagram is drawn in Fig. 10(d)~(f), and the vertical axis of the figure is the absolute error between the predicted value of SOH and the actual value. As shown in Fig. 10(d)~(f), the model proposed in this study not only has the smallest mean error in the three cases but also has the most concentrated error range, which shows that CNN-LSTM-Skip has the best performance in SOH prediction. The detailed error data of the model is shown in Table 6, Table 7, and Table 8. When using 70% cycle data to train the model, ARMSE has improved by 60% compared with CNN-ASTLSTM. In the other two cases, the CNN-LSTM-Skip model still has relatively high accuracy, and the coefficient of determination is both above 0.9. The above results

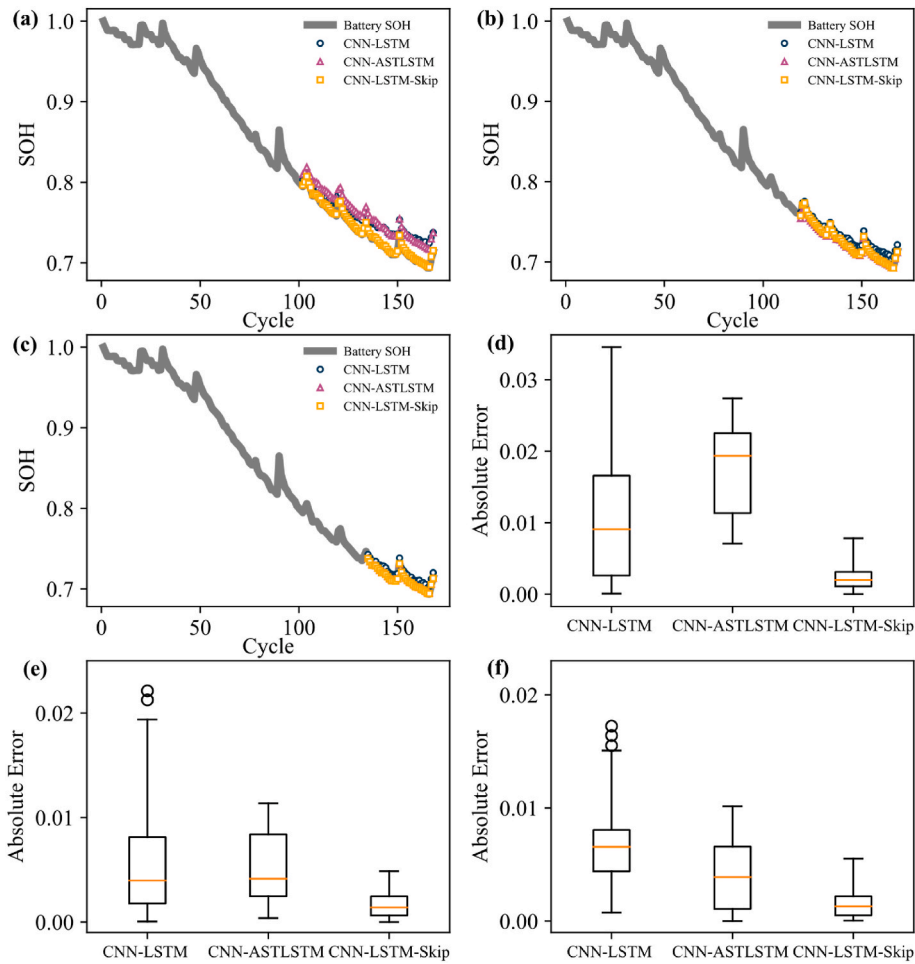


Fig. 10. (a)~(c) are the prediction results of NASA B0005 using 60%, 70%, and 80% cycle data respectively, (d)~(f) are the prediction results of all batteries using 60%, 70%, and 80% cycle data box plot of forecast absolute error.

Table 6
Prediction results of different models using 60% cycle data training.

Methods	battery ID	R^2	MAE	ARMSE
CNN-LSTM	B0005	0.5965	0.0184	0.0121
	B0006	0.8467	0.0123	
	B0007	0.9932	0.0016	
	B0018	0.5781	0.0109	
CNN-ASTLSTM	B0005	0.5074	0.0213	0.0179
	B0006	0.9404	0.0094	
	B0007	0.3526	0.0200	
	B0018	-0.2649	0.0201	
CNN-LSTM-Skip	B0005	0.9960	0.0017	0.0025
	B0006	0.9946	0.0023	
	B0007	0.9795	0.0033	
	B0018	0.9933	0.0012	

Table 7
Prediction results of different models trained with 70% cycle data.

Methods	battery ID	R^2	MAE	ARMSE
CNN-LSTM	B0005	0.8819	0.0065	0.0064
	B0006	0.9039	0.0077	
	B0007	0.9884	0.0016	
	B0018	0.8343	0.0052	
CNN-ASTLSTM	B0005	0.9832	0.0024	0.0054
	B0006	0.9122	0.0096	
	B0007	0.9567	0.0035	
	B0018	0.8738	0.0048	
CNN-LSTM-Skip	B0005	0.9976	0.0009	0.0019
	B0006	0.9949	0.0021	
	B0007	0.9938	0.0012	
	B0018	0.9662	0.0024	

show that the CNN-LSTM-Skip model is more accurate and has better robustness in different situations.

4.2.2. The experiment with the data from oxford

Fig. 11(a)~(c) are the prediction results of three of the batteries, indicating the effectiveness of the model proposed in this work. Fig. 11 (d)~(f) are error box diagrams. First, in Fig. 11(b), the SOH degradation patterns are different from the other batteries, which is due to a sudden destructive event in the positive electrode [37]. From Fig. 11, we can conclude that the proposed model has the smallest error mean and the most concentrated range. Table 9 details the results of the training using

60%, 70%, and 80% of the cycle data and shows the average of the prediction results for the eight cells. The results indicate that the coefficient of determination value of the model proposed in this work is above 0.95 in all three cases, and the RMSE is within 0.4%. Compared with other models, the RMSE of the CNN-LSTM-Skip model is improved more than 60%. These results illustrate that the CNN-LSTM-Skip model has a better effect on the SOH prediction task and maintains high accuracy in different situations.

Table 8

Prediction results using 80% cycle data training.

Methods	battery ID	R^2	MAE	ARMSE
CNN-LSTM	B0005	0.6727	0.0093	0.0074
	B0006	0.8439	0.0134	
	B0007	0.7515	0.0072	
	B0018	0.7624	0.0092	
CNN-ASTLSTM	B0005	0.9934	0.0008	0.0043
	B0006	0.9105	0.0071	
	B0007	0.8763	0.0035	
	B0018	0.8979	0.0044	
CNN-LSTM-Skip	B0005	0.9966	0.0006	0.0017
	B0006	0.9961	0.0013	
	B0007	0.9162	0.0031	
	B0018	0.9921	0.0012	

4.3. Model performance comparison

The model performance data is collected from the training process of the NASA dataset using 60% of the data. The model performance comparison is shown in Table 10. From the perspective of performance, the CNN-LSTM-Skip model increases the amount of calculation and calculation time and has no advantage over CNN-ASTLSTM and CNN-LSTM, but the calculation cost is also within an acceptable range.

5. Conclusion

In this study, the CNN-LSTM with skip connection method is proposed for SOH estimation of LIBs and verified on different datasets. The

main conclusions of our approach are as follows:

1) To reduce the input data of neural networks, this work uses a variable importance algorithm to calculate the contribution of input features to model output. By employing the variable importance algorithm, the amount of input data is reduced by 40% on the NASA dataset and 20% on the Oxford dataset. This work also uses the IC curve as an input feature and verifies the effectiveness of IC curves in SOH

Table 9

Results of the Oxford dataset.

Cycle data percentage	Methods	RMSE	R^2	MAE
60%	CNN-LSTM	0.0117	0.7785	0.0099
	CNN-ASTLSTM	0.0105	0.8059	0.0091
	CNN-LSTM-Skip	0.0037	0.9600	0.0033
70%	CNN-LSTM	0.0101	0.5952	0.0090
	CNN-ASTLSTM	0.0069	0.7983	0.0061
	CNN-LSTM-Skip	0.0018	0.9757	0.0015
80%	CNN-LSTM	0.0083	0.5072	0.0079
	CNN-ASTLSTM	0.0032	0.9063	0.0028
	CNN-LSTM-Skip	0.0013	0.9729	0.0011

Table 10

Model performance comparison results of the NASA dataset.

Model	FLOPs	Parameter	Storage (KB)	Time(s)
CNN-LSTM	752,284	42,101	532	78.2366
CNN-ASTLSTM	716,593	43,101	534	88.3997
CNN-LSTM-Skip	1,113,227	62,101	784	124.8946

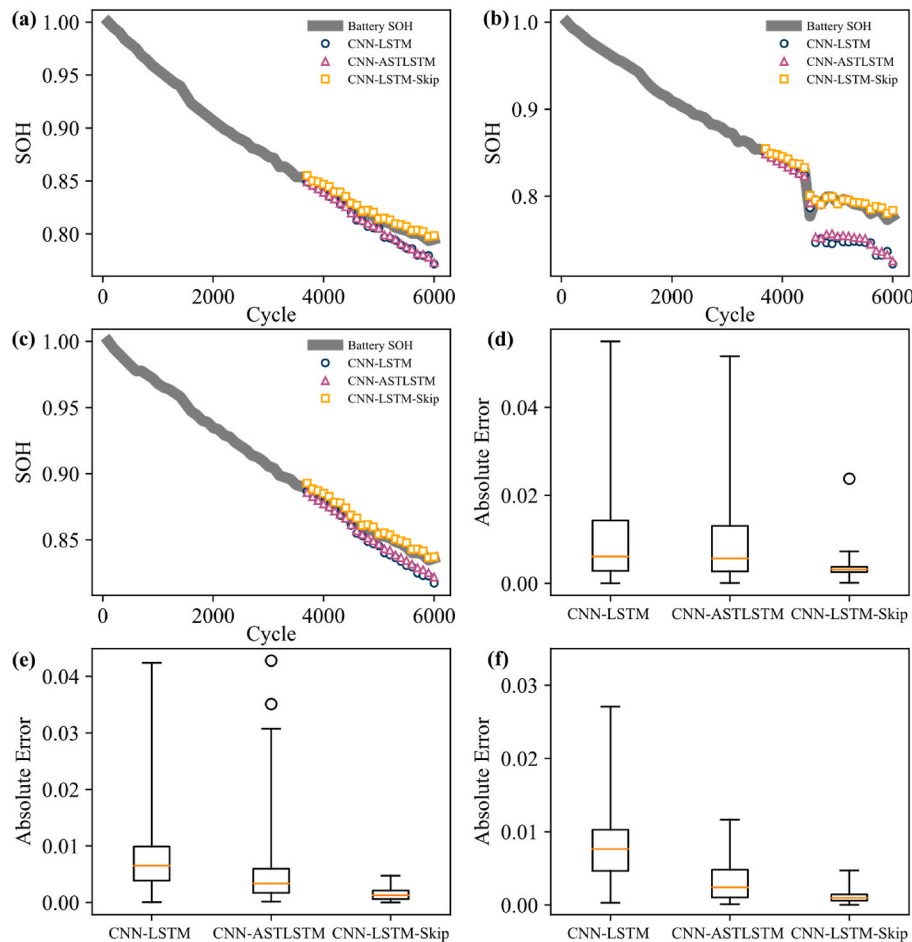


Fig. 11. (a)~(c) are the prediction results of the three batteries trained by cattle using 60% cycle data, and (d)~(f) are using 60%, 70%, and 80% cycle data respectively. Boxplot of absolute errors for all battery predictions.

prediction.

- 2) This study proposes a SOH prediction model based on the CNN-LSTM with skip connection. The skip connection is added to the CNN-LSTM to improve the training effect and address the degradation problem of multilayer stacked LSTM.
- 3) The CNN-LSTM, CNN-ASTLSTM, and CNN-LSTM-Skip models are compared, and 60%, 70%, and 80% cycle data are used for SOH prediction on the NASA dataset and the Oxford dataset, respectively. The proposed model can well perform SOH estimation in all three cases, and the RMSE is less than 0.004. From the perspective of prediction result accuracy, the CNN-LSTM-Skip model has higher prediction accuracy on both datasets. Compared with the other models, the proposed model can improve the RMSE by more than 40% in most cases, the average error value is lower, and the error distribution range is smaller.

Overall, the proposed method achieves accurate SOH estimation in different cases and the model is more applicable to different datasets. In the following work, the model will be simplified, and the consumption of computational resources will be reduced under the premise of ensuring the accuracy of prediction.

Credit author statement

Huanwei Xu: Writing, Conceptualization, Visualization. **Lingfeng Wu:** Writing, Methodology, Experiments. **Shizhe Xiong:** Experiments, Writing. **Wei Li:** Supervision, Methodology, Writing - review & editing. **Akhil Garg:** Supervision, Writing. **Liang Gao:** Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data availability

Data will be made available on request.

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